

Next-Stage Research Plan: Deep Reading, Methodology Alternatives, and Experiments

A faithful PDF edition of the plan developed after reviewing the working research proposal and prior literature-review discussions.

Your proposed approach - reading the 13 closest papers while progressively designing alternative solutions - is appropriate. However, I recommend one important change: **we should organize the next stage around research questions, competing methodological alternatives, and small experiments, rather than simply finishing each reading phase and designing its corresponding module.**

This will help us avoid spending weeks designing an elaborate tracking and Re-ID pipeline before discovering whether our central contribution, sequential evidence acquisition, actually improves recovery.

1. Where our research currently stands

Reliability-Aware Sequential Identity Evidence Acquisition

Our task is to track one user-selected road vehicle and recover the same physical vehicle after occlusion or disappearance, including when visually similar distractors appear.

Our central research question: Can reliability-aware sequential acquisition of vehicle identity cues improve post-occlusion recovery among visually similar distractors while reducing false identity decisions and average inference cost compared with always-on multimodal Re-ID and fixed coarse-to-fine recovery?

Literature screening: 26 papers. **Selected for deep reading:** 13 papers.

The central novelty is not tracking, multimodal vehicle identification, or a cheap-first cascade by itself. It is the specific recovery decision mechanism: deciding what existing evidence means, whether it is reliable, which additional cue to acquire, and when to accept, reject, or remain uncertain. The proposal explicitly leaves the mathematical estimator open to comparison.

2. A more effective next-stage research plan

I suggest five interconnected phases, with one additional preliminary task.

Phase 0 - Define the recovery experiment first

Establish a small evaluation set, event definitions, and cost measurements before committing to algorithms.

Phase 1 - Tracking, occlusion and candidate generation

Read four papers and identify a suitable baseline tracker, loss detector, and recovery candidate generator.

Phase 2 - Vehicle identity and evidence representation

Read three papers and define candidate identity cues, target memory, reliability, and missing observations.

Phase 3 - Sequential evidence acquisition

Read four papers and develop alternative mathematical formulations for belief updating and next-cue selection.

Phase 4 - Decision-making and uncertainty

Read the remaining two papers and formulate acceptance, rejection, abstention, and stopping decisions.

Phase 5 - Integrated prototypes and comparative experiments

Connect the modules, compare alternative solutions, perform ablations, and refine the contribution.

The reason for adding Phase 0 is that our evaluation is event-level, not standard Re-ID retrieval alone. An event begins when target identity becomes uncertain and involves candidates that may or may not contain the true target. We should establish a small set of such events before designing sophisticated policies.

For example, collect cases featuring a readable plate, an unreadable plate, nearly identical vehicles, a substantial viewpoint change, and target absence after occlusion. Record the ground-truth identity and available cues, then establish what information and computational costs a policy would actually encounter.

3. Reading the papers and developing our methodology

Your categorization is useful, and I would retain its three groups while adding a fourth.

Phase 1 - Tracking, occlusion and recovery | 4 papers

Understand how existing trackers preserve identity, identify target loss, handle distractors, and generate reappearance candidates.

Paper	What we need to extract
VeTrack	Appearance representation and confidence-guided association
Sentinel	Track uncertainty and prolonged occlusion handling
DAM4SAM	Distractor-aware memory and target recovery
DeconfuseTrack	Mechanisms for reducing confusion between target and distractor

Methodology deliverable: a tracker-independent recovery interface. A target selection produces initial memory; ordinary tracking continues while reliable; a loss or identity-ambiguity trigger starts recovery and generates candidates.

Instead of selecting one complete paper architecture, we record alternatives for loss detection, memory updating, and candidate generation. These alternatives must expose compatible inputs and outputs so we can exchange them during testing.

Phase 2 - Vehicle identity representation | 3 papers

Understand how persistent identity is represented when some cues are weak, degraded, unavailable, or contradictory.

Paper	What we need to extract
Dual License Plate + Visual Identification	Plate and appearance representation and combination
Xu et al. - Coarse-to-Fine Vehicle Re-ID	Attribute and appearance retrieval stages
SAD-ReID	Semantic identity and memory reliability

Methodology deliverable: a common evidence representation for plate, color, body type, make/model, visual embedding, and local details.

Each observation should specify its identity evidence, measured reliability, availability, and acquisition cost. Missing plate information must not be interpreted as a different plate, and several highly correlated cues must not automatically count as independent confirmation.

Phase 3 - Sequential evidence acquisition | 4 papers

This phase develops the candidate mathematical methods for our main contribution.

Paper	What we need to extract
Active Feature Acquisition Survey (2026)	Alternative acquisition objectives and policy families
Ji & Carin (2007)	Sequential decisions and information-based acquisition
Sequential Cost-Sensitive Feature Acquisition (2016)	Incorporating feature costs into sequential decisions
Cost-Aware Multimodal Person Re-ID (2026)	Conditional expensive-modality execution and early exit

Methodology deliverable: several explicit acquisition policies that operate on the same identity evidence and are tested under the same computational budget.

We can start with fixed-order and reliability/cost heuristics, then investigate expected information gain, decision-risk reduction, and possibly a learned policy.

Phase 4 - Reliability, rejection and stopping | 2 papers

These are the two papers missing from your current three-phase categorization.

Paper	What we need to extract
Trust, but Verify - Uncertainty-Driven Evidential Multimodal Learning	Treatment of conflicting evidence and uncertainty
Machine Learning with a Reject Option - A Survey	Selective decisions, abstention, and decision risks

Methodology deliverable: a formally specified MATCH / REJECT / NEED MORE EVIDENCE decision mechanism, with an additional unresolved outcome if the evidence budget is exhausted.

The fourth phase is particularly important. A next-cue acquisition policy needs a definition of what constitutes sufficient evidence; otherwise, uncertainty reduction alone may continue requesting cues without improving the actual match/reject decision.

4. How we should document alternative solutions

For each module, we should maintain a methodology decision register. This builds on your Formula 2.A / 2.B idea, but makes every alternative testable.

Example: Module 3 - Next-cue selection

Illustrative hypotheses to investigate, not finalized algorithms.

Variant	Proposed mechanism
3.A	Fixed cue order with confidence thresholds
3.B	Hand-designed reliability-to-cost priority
3.C	Expected information gain divided by acquisition cost
3.D	Expected reduction in identity decision risk minus acquisition cost
3.E	Learned sequential acquisition policy, if justified by pilot experiments

Example mathematical candidates

Information-gain formulation: $k^* = \operatorname{argmax}_k E[\Delta H_k] / c_k$

Decision-risk formulation: $k^* = \operatorname{argmax}_k (E[\Delta R_k] - \lambda c_k)$

Here H denotes identity uncertainty, R denotes decision risk, c_k is the cost of acquiring cue k , and λ controls the cost trade-off. Availability constraints and the treatment of reliability still need to be specified.

Every proposed variant should have a record containing its source paper, assumptions, equations, input/output requirements, expected advantage, likely failure conditions, additional training needs, and experiment results. We should distinguish a formula directly reproduced from a paper from our own adaptation or new proposal.

One especially useful addition is a decision log: after each paper, record whether it changes our existing methodology, introduces a new competing variant, or simply provides a baseline. This prevents the 13-paper reading exercise from becoming 13 disconnected summaries.

5. Experiments should begin before all 13 papers are finished

I would not wait until Phase 4 to run our first experiment. Once we have a basic tracker, a small recovery-event dataset, and a simple identity representation, we can establish tracker-only and always-on Re-ID results.

As we read the sequential-acquisition papers, we should first compare policies using precomputed cue outputs. However, these outputs must come from realistic post-occlusion observations, and measured model-invocation costs must be included. This enables rapid policy comparisons without repeatedly running every identity model, while avoiding the false conclusion that precomputed features are free at deployment.

We then move promising policies into an integrated system and measure their actual latency.

The proposal identifies six comparison systems; our primary methodological comparisons should include the following:

Comparison	Research purpose
Always-on multimodal fusion	Establish recovery quality when every available cue is acquired
Fixed coarse-to-fine	Test against our previous hierarchical approach
Fixed-order confidence cascade	Isolate the value of dynamic cue ordering
Sequential selection without explicit reliability	Isolate reliability modeling
Reliability-aware sequential selection	Evaluate the combined proposed mechanism

The main outcomes should be correct recovery, false recovery, false rejection, time to recovery, calibration, and inference cost. Reporting accuracy against cost across several operating budgets will be more informative than claiming success from one threshold setting.

There are three methodological safeguards I would establish now. First, a cue's reliability must be estimated without knowing the ground-truth candidate identity at test time. Second, tuning of acceptance/rejection thresholds and cost weights must occur on separate development data, not our final test events. Third, when the system rejects every currently visible candidate, we must distinguish that decision from permanently abandoning the selected target: the correct vehicle may simply not have reappeared yet.

6. The immediate next step

We should start with **Phase 0 and the first Phase 1 paper, VeTrack**, using a consistent deep-reading template. For every paper, we will extract the actual algorithm and equations, identify what is transferable to our task, create or revise specific methodology variants, and define a small experiment that could distinguish those alternatives.

At the end of each phase, we will produce two documents: an updated methodology decision register and a compact experimental specification. We will not freeze a solution just because it appears in a published paper or looks mathematically elegant.

This preserves your idea of progressively building Solution A with alternatives 1.A, 1.B, 2.A, and so forth, while making experimental evidence - not reading order - the deciding factor.

Source note: Plan based on the uploaded Finalized Working Research Proposal v2 (September 2026), especially pages 2-5 and 8-11, and the project literature-review discussion.